Blank / Junk — Methodology

Status: **IMPLEMENTED (2026-10-15)** in `core-pipeline/stages/lib/blank_junk/method.py`, called by the blank/junk stage in both passes. Written 2026-10-14; R4 narrowed on 2026-10-15 (§6). Canon: `core-pipeline/stages/lib/keyword-canon/blank_junk_context_canon.json`.

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1. Principle

A clinical page that is junked loses billable content and cannot be recovered downstream. A junk page that is kept only costs a reviewer a few seconds. So:

- Every tie, abstention and missing signal resolves to **KEEP**.
- JUNK needs **agreement**: at least two independent methods saying JUNK and no veto.
- A small set of deterministic rules decide before any vote, so obvious pages never depend on a model.
- Every vote is written to the database, so any verdict can be explained and thresholds can be tuned from data.

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2. Verdicts

Verdict	Meaning	Subtypes
`blank`	No content	—
`junk`	Non-clinical page	Invoice, Cover Page, Record Request/Transmittal, Instructions, Letter/Fax, Others
`duplicate`	Same content as a kept page nearby	—
`not_blank_junk`	Main page, kept	—

A junk verdict always carries a subtype. Others is the fallback; NULL is never written.

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3. Inputs

Signal	Source	Pass 1	Pass 2
Text	Tesseract prelim (pass 1); Azure DocIntel, else Docling/RapidOCR (pass 2)	yes	yes
Line coordinates	`final2.pagesMeta[].lines[].polygon`, else `final1.words[].polygon`	no	yes
Ink map	Page image (same ink measure the HW stage uses, `page_marks`)	yes	yes
Page type	`ocr_quality_results.document_type`: printed / handwritten / form / **visual** / blank / uncertain, and `handwritten_area_pct`, page-tag model (`convnext_page_tags`)	yes	yes
Signature	KV extraction (`kv_extract`, seq 56): Provider Signature = Yes. Runs after final OCR and **before** pass 2 (seq 60)	no	yes
DistilBERT	8-label page model (KEEP, BLANK, 6 × JUNK_*)	yes	yes
TF-IDF	`tfidf_flat.joblib`, p(junk)	yes	yes
Keyword context	`blank_junk_context_canon.json`	yes	yes

Visual. Read from one place only: `ocr_quality_results.document_type = 'visual'` (CSV: `document_type` in `imaging/<chart>_hw_printed.csv`). This is the same value review-ui shows as Type. No extra confidence cut-off is applied: the page-tag model already turns a weak call into `uncertain`. There is no proxy. A page without page-tag output (`method` other than `convnext_page_tags`) is simply not Visual. Its row records `visual_source = "none"`, so a run on the old 2-class checkpoint shows in the data. The four charts under `review-ui/data/folders/` were run that way and carry no Visual tags.

A Visual page is decided in **pass 1** (R8) and **pass 2 is skipped** for it.

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4. Context canon (the JSON)

`blank_junk_context_canon.json` holds lists of words. It reloads on change, like the other canons. The words are sorted into three **contexts**, and each context is split into **groups** of similar words:

Context	Groups	What a high score means
clinical	section_headings, assessment_language, vitals, exam_findings, diagnosis_coding, conditions, medications_orders, labs_results, imaging_procedures, **patient_education**, signature	The page is part of a medical record
junk	fax_cover, cover_page, cover_stamps (Accept / Unaccept), record_request, roi_vendors, invoice, placeholder	The page is retrieval or billing paperwork; each group names the junk subtype
demographic	identity, registration	A face sheet or registration page: administrative-looking, but a chart page
neutral	fax, phone, page, confidential, PHI, and all insurance wording	Never scored: it appears on every kind of page

Two lists are deliberately absent:

- **Patient education counts as clinical.** Handouts and instruction sheets (“Consumer Information Use and Disclaimer”, “after visit summary”, “call your doctor if”) are part of the record. Page classification has a Patient Education page type, but it runs after blank/junk, so a junked handout never reaches it. Two `patient_education` hits make the page C, so DistilBERT cannot force it to junk (R4) and the vote keeps it.
- **No `instructions` junk group.** Patient instructions (discharge, medication, after-visit) are clinical pages. The Instructions subtype can still come from DistilBERT, but no word list pushes a page toward it.
- **No `insurance` group.** Insurance wording sits on clinical notes, face sheets and retrieval letters alike, so it tells us nothing. It is neutral.

4.1 How a page gets a score

Each group has two numbers.

- **Weight:** how much one hit of that group is worth. It reflects how sure the word makes us.
- **Cap:** the most the group can add in total, however many times its words appear.

Weight	Meaning	Example
1	Common or ambiguous: suggests, never proves	“diabetes”, “mg”, “dob”: also appear in request letters and forms
2	Typical of the context	“Assessment:” at line start, “fax cover sheet”, “face sheet”
3	Nearly decisive	“request for medical records”, “invoice number”, “electronically signed by”
4	Decisive on its own	An ROI vendor name: Datavant, Ciox, Optum
6	Placeholder only	“intentionally left blank”

The calculation, in three steps:

1. **Count unique hits.** Each distinct word or phrase of the group counts **once**, however often it appears. “Datavant” on every line of a letterhead is 1 hit; “Datavant” and “Ciox” are 2. Where phrases overlap, the longest wins: “fax cover sheet” is one hit, not also a hit for “fax cover”.
2. **Group score** = weight × unique hits, cut down to the cap if it is higher.
3. **Context score** = the group scores of that context added together.

The page ends up with three numbers: **clinical**, **junk** and **demographic**. Each is then compared with its threshold:

Threshold	Value	Name used below
clinical ≥ 4	4	**C**: clinical evidence present
clinical ≥ 8	8	**C⁺**: clearly a clinical note
junk ≥ 4	4	**J**: junk evidence present
junk ≥ 7	7	**J⁺**: clearly retrieval / billing paperwork
demographic ≥ 3	3	**D**: demographic page

Worked example 1 — a progress note.

Group	Unique hits on the page	Weight	Weight × hits	Cap	Group score
section_headings	“HPI:”, “Assessment:”, “Plan:” = 3	2	6	8	6
conditions	“diabetes” (3 times, counts once), “hypertension” = 2	1	2	4	2
medications_orders	“metformin”, “lisinopril”, “daily”, a dose like “500 mg” (the dose pattern counts once) = 4	1	4	3	**3** (capped)
				clinical	**11 → C⁺**

Worked example 2 — a record-request letter that lists diagnoses.

Group	Unique hits	Weight	Score
record_request	“request for medical records”, “dates of service requested” = 2	3	6
roi_vendors	“Datavant” (twice, counts once) = 1	4	4
		junk	**10 → J⁺**
conditions	“diabetes”, “CKD” = 2	1	2
		clinical	**2: below C**

Why unique hits. A word repeated in a letterhead, a footer or a fax banner on every line says nothing new after the first time. Counting it once keeps repetition from building a score.

Why the cap matters. A medication list naming 20 different drugs has 20 unique hits; without a cap it would score 20 and look like the strongest clinical page in the chart. With the cap of 3, one kind of word can never reach a threshold alone. For example, conditions max 4 < C⁺ at 8. A page reaches C⁺ only with evidence from **several different groups**: headings, plus conditions, plus medications. Junk works the same way. One vendor name gives at most 8 points; a vendor plus request wording is J⁺. The fax-cover group is capped at 6, so a fax cover alone is J but never J⁺.

Three refinements:

- `line_start` (section headings only): “Plan” counts only at the start of a line, alone or followed by `:` or `–`. Inside a sentence (“we plan to…”) it does not count.
- **Patterns** catch value-bearing evidence: `BP 128/76`, `10 mg`, `$45.00`. A form label “BP: _____” does not match; a measured BP does.
- **Zones** (pass 2 only): the header is the top 12% of the page, the footer the bottom 8%. Junk words there count only if the body also has a junk hit. A fax banner across the top of a lab report does not make it junk.

5. The three voters

Each voter returns **JUNK**, **KEEP** or **ABSTAIN**, a strength (**strong** / **weak**) and, for JUNK, a subtype. The voters are independent: none reads another's output.

5.1 DistilBERT (semantic)

The raw softmax over the 8 labels is used, not the vendored decision rules. Those rules embed their own vetoes, which would count the same evidence twice.

```
p_keep  = P(KEEP)
p_junk  = Σ P(JUNK_∗) + P(BLANK)        # BLANK on a page with text is junk
top_junk_label = argmax over JUNK_∗ (BLANK → Others)
```

Condition	Vote	Strength
fewer than 8 words of text	ABSTAIN	—
p_junk ≥ 0.85	JUNK	strong
0.60 ≤ p_junk < 0.85	JUNK	weak
p_keep ≥ 0.85	KEEP	strong
0.50 ≤ p_keep < 0.85	KEEP	weak
otherwise	ABSTAIN	—

Subtype: `top_junk_label`.

5.2 TF-IDF (lexical statistics)

Condition	Vote	Strength
p_junk ≥ 0.85	JUNK	strong
0.60 ≤ p_junk < 0.85	JUNK	weak
p_junk ≤ 0.15	KEEP	strong
0.15 < p_junk ≤ 0.40	KEEP	weak
0.40 < p_junk < 0.60	ABSTAIN	—

Subtype: none of its own. It agrees or disagrees on JUNK, and the subtype comes from the others (§8).

5.3 Keyword context (canon)

J	C	D	Vote	Strength	Subtype
J^+	—	—	JUNK	strong	best junk group
J	—	—	JUNK	weak	best junk group
J / J^+	—	D	JUNK if the junk comes from record_request / roi_vendors / invoice, else KEEP (weak)	as above	best junk group
any	C	any	KEEP	weak	—
any	C^+	any	KEEP	strong	—
—	—	D	KEEP	weak	—
—	—	—	ABSTAIN	—	—

“—” means below threshold. When J and C both hit, the keyword voter says KEEP and the call is uncertain (§10). The best junk group is the group with the highest score; a tie goes to the group with the higher weight.

The keyword voter **abstains** when it has no evidence; it does not vote KEEP. A page with no hits is decided by the two models.

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6. Precedence ladder

Rules run top to bottom. The first rule that fires decides, and nothing below it runs. Rules R1–R8 are deterministic; R9 is the vote.

#	Rule	Verdict	Overrides
R1	**Blank.** No ink above the blank threshold (`blank_page` / `faint_marks_only` from the HW stage), **or** fewer than 15 non-space OCR characters and the page is not Visual	`blank`	everything
R2	**Placeholder.** A `placeholder` term ("intentionally left blank") and 25 words or fewer	`junk` / Others	everything below
R3	**Signature.** Provider Signature = Yes from KV extraction (`kv_extract`, seq 56, runs before pass 2), or one of the clinical `signature` group's phrases ("electronically signed by", "attestation" …; a credential at a line end alone is not enough)	`not_blank_junk`	R4–R9
R4	**DistilBERT force.** DistilBERT p(junk) **> 0.85** (Σ JUNK_* + BLANK), pass 2, **unless the canon finds clinical (C, ≥ 4) or demographic (D, ≥ 3) evidence** — then the page goes on down the ladder and the vote decides (`votes.r4_skipped`)	`junk` / DistilBERT's top junk label	R5–R9
R5	**Short page.** Content ends in the top **15%** of page height (§7). Fires **even when the page carries a clinical heading**, and **whatever the neighbouring pages are**: no adjacency check	`junk` / Others	R6–R9, including the clinical veto
R6	**Junk counter-veto.** record_request + roi_vendors ≥ 7 (J^+ from retrieval terms alone) and not C^+	`junk` / Record Request/Transmittal	R7–R9
R7	**Strong clinical.** C^+ (score ≥ 8)	`not_blank_junk`	R8–R9
R8	**Visual.** `document_type = 'visual'`. Decided in **pass 1**; pass 2 is skipped for these pages	`not_blank_junk`, unless invoice / record_request / roi_vendors ≥ 7, then `junk` with that group's subtype	R9
R9	**Vote** (§8)	per §8	—

Why this order:

- **DistilBERT force (R4).** Above 85% the model is trusted outright, except on a page the canon calls clinical or demographic. Run on the local charts, the unrestricted rule junked a Patient Demographics face sheet and a BCG medication-administration page (DistilBERT p(junk) 0.98–1.0 on both), so on 2026-10-15 it was narrowed: those pages go to the vote, where the clinical and demographic vetoes apply. Only blank, placeholder and signature sit above R4. A signed page stays kept, for the same reason it is exempt from the short-page rule (next point).
- **Signature (R3) above DistilBERT force (R4) and short page (R5).** In risk adjustment a diagnosis is only valid on a signed, credentialed note. A page holding only “Electronically signed by Dr X, MD” at the top is the attestation of the previous page’s note. Junking it can invalidate that date of service. This is the **only** exemption from R5.
- **Short page (R5) above the clinical veto.** As requested: a lone “Assessment and Plan” heading or a header block with nothing beneath it is not codable content.
- **R6 above strong-clinical (R7) only below C⁺.** A request letter lists diagnoses and dates (“please send records for diabetes, CKD”), which gives a moderate clinical score. A real note reaches C⁺ and survives R6.
- **Visual (R8)** has little text, so the text voters are unreliable on it. It is kept by default; only very strong retrieval or invoice wording junks it. A logo-heavy fax cover can be tagged Visual.

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7. Short page (R5) — exact definition

Pass 2 only, because it needs line coordinates and Provider Signature from KV extraction (which runs just before pass 2).

1. **Page height H:** `pagesMeta.height` (DocIntel), else `final1.height`. Normalise every polygon to top-left origin, since Docling headers may be bottom-left.
2. **Lines:** DocIntel lines; else Docling words grouped into lines (same baseline ± half the median word height).
3. **Drop** lines that start in the bottom 6% (page numbers, footers) and lines matching `ignore_line_patterns` **anywhere in the line**: “Page 2 of 5” / “Pg 14 of 17”, a bare number, “Printed: ”, fax transmission stamps, “Confidential…”. A running footer such as `MRN: 023466 Printed: 10/30/2025 8:10 AM Pg 14 of 17` at 92% of the height is a stamp, not content.
4. **content_bottom** = max bottom-y of the remaining lines ÷ H.
5. **Ink check:** the band from 17% to 94% of the height must hold no ink above `body_ink_max_fraction` (1e-4, the same blank cut-off as the HW stage). This catches a picture, a handwritten block or a stamp that OCR did not read.
6. **Fire** when `content_bottom ≤ 0.15` and the ink check passes and R3 did not fire.

Not considered:

- Clinical headings on the page.
- The previous and next page.
- Word count: a page that ends in the top 15% is short by construction.

Page	Result
“PROGRESS NOTE — Patient: J Doe DOB 01/02/1950”, nothing below	JUNK / Others
“Assessment and Plan:” alone at the top	JUNK / Others
Header block, then a wound photo mid-page (no text)	Kept: ink in the body band
Header block, then “Page 3 of 7” at the bottom	JUNK / Others: footer ignored
“Electronically signed by A. Smith, MD 03/14/24” alone at the top	Kept by R3

8. How the three voters agree (R9)

8.1 Counting

Each voter casts one ballot: J (JUNK), K (KEEP) or A (ABSTAIN). Strong and weak are kept on every ballot. With three voters there are ten possible ballot sets.

8.2 Decision table

`s` = strong, `w` = weak. “C” = the keyword clinical score is at or above 4.

#	Ballots	Verdict	Uncertain (confidence ≤ 0.60)	Notes
1	J J J	**JUNK**	no; yes if any J is weak or C	Unanimous
2	J J K	**JUNK** only if both J are strong **and** the K is weak; else **KEEP**	yes	One dissent. A strong KEEP blocks
3	J J A	**JUNK** if at least one J is strong; else **KEEP**	yes if any J is weak	Two agree, one silent
4	J K K	**KEEP**	yes if the J is strong	Majority keep
5	J K A	**KEEP**	yes	Split, no majority
6	J A A	**KEEP**	yes if the J is strong	One voice is not agreement
7	K K K	**KEEP**	no	
8	K K A	**KEEP**	no	
9	K A A	**KEEP**	no	
10	A A A	**KEEP**	yes	Nothing could judge the page

8.3 Vetoes applied to a JUNK result from the table

Applied in order to rows 1–3. Any veto turns the result into **KEEP, uncertain**.

Veto	Condition	Effect
Clinical veto	C (clinical ≥ 4)	KEEP, uncertain. Even 3 strong JUNK ballots are blocked
Demographic veto	D and the junk score does not come from record_request / roi_vendors / invoice	KEEP, uncertain (face sheets, registration)
Handwritten veto	`handwritten_area_pct` **> 30%** (page-tag model)	KEEP, uncertain. A page that is mostly handwriting is clinical far more often than admin, and OCR on it is weak. A printed form with a few handwritten fields (≤ 30%) is not protected

C⁺ never reaches R9 (R7 kept it). So the clinical veto here covers the moderate band, score 4–7.

8.4 Why two-of-three, and why strength matters

- **Independence.** DistilBERT reads meaning, TF-IDF reads word statistics, and the canon reads curated phrases. They fail differently: BERT is fooled by long letters, TF-IDF by vocabulary overlap, and keywords by OCR noise. Agreement between two of them is much rarer by chance than either alone.
- **A strong KEEP is a veto in all but name** (row 2). If any method is confident the page is clinical, two other methods saying junk is not enough.
- **Two weak JUNKs are not agreement** (row 3 with both weak, and row 2). Weak means “leaning”. Two leanings plus a silent third stays KEEP, with low confidence.
- **Abstention is not a KEEP vote.** It removes the voter. That is why row 6 (one J, two silent) keeps the page: the only evidence is one voter.

8.5 Confidence written to the row

```
JUNK:  confidence = mean(p_junk of the voters that voted JUNK)
       (keyword p = min(1, junk_score / 10))
KEEP:  confidence = mean(1 − p_junk of the voters that voted KEEP), or 0.50 if none did
Rules R1–R8: fixed values — R1 1.00, R2 0.99, R3 0.95, R5 0.95, R6 0.97, R7 0.95, R8 0.90; R4 =
DistilBERT's p(junk)
```

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9. Subtype

When the verdict is JUNK (R9), the subtype is chosen in this order:

1. The keyword voter’s best junk group, if the keyword voter voted JUNK.
2. Else DistilBERT’s `top_junk_label`, if BERT voted JUNK.
3. Else Others.

The call is uncertain when keyword and BERT both voted JUNK with **different** subtypes and both were strong.

Mapping: fax_cover → Letter/Fax · cover_page → Cover Page · record_request, roi_vendors → Record Request/Transmittal · invoice → Invoice · placeholder → Others · cover_stamps → Cover Page. There is no instructions word list, so Instructions comes only from DistilBERT’s label.

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10. Uncertain calls: lower confidence, never “no verdict”

This is client-facing, so every page gets one verdict: blank, junk (with a subtype), duplicate or Main. There is no separate review flag. When the call is uncertain, the verdict stands and its **confidence is capped at 0.60** (`uncertain_confidence_cap` in the canon). Filter on confidence ≤ 0.60 to see the pages worth a second look.

A call is uncertain when:

- the table in §8.2 marks it,
- any veto in §8.3 fired,
- J and C both hit,
- subtypes disagree (§9),
- a voter was unavailable (§12).

Rules R1–R8 are not uncertain: they keep their fixed confidence.

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11. Pass 1 and pass 2

	Pass 1 (Tesseract)	**Pass 2 (final OCR)**
Pages	All pages. Visual is **decided here** (R8). Handwritten and uncertain pages are deferred	Everything pass 1 deferred; **Visual pages skipped**
Rules	R1, R2, R6, R7, R8	R1–R9 in full (R8 never reached: Visual is already decided)
DistilBERT > 0.85	Not forced (no Provider Signature yet); counts as a strong JUNK ballot	**Forced JUNK** (R4) unless signed
Vote	Only **row 1 with all three strong** and no veto is final	Full table
Everything else	Deferred: stays Main, gets final OCR, decided in pass 2	Final

Pass 1 only removes pages it is certain about, which saves the stage-5 bill. A page that pass 1 junks never gets final OCR, so pass 1 must never be wrong cheaply. DistilBERT force (R4), short page (R5) and signature (R3) wait for pass 2 because they need coordinates and Provider Signature, which KV extraction writes just before pass 2.

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12. Degraded runs

Missing	Behaviour	Recorded
DistilBERT	Its ballot is A. Two voters remain, so JUNK needs both, and both strong	`"note": "unavailable"` on the DistilBERT ballot; confidence ≤ 0.60
TF-IDF	Same	`"note": "unavailable"` on the TF-IDF ballot; confidence ≤ 0.60
Canon fails to load	Last good version (canon_store); first load raises	—
Page-tag model	No page is Visual; R8 never fires	`visual_source = "none"`
KV extraction (Provider Signature)	R3 uses the canon `signature` group only	`signature_source = "canon_only"`
Handwritten area (old 2-class model)	Handwritten veto does not apply	`hw_area = "none"`
Coordinates (pass 2 text only)	R5 does not run	`short_page = "no_geometry"`

13. Duplicates

Run after the verdict, on kept pages only. The method is unchanged: normalised text of at least 300 characters, SequenceMatcher ≥ 0.98 or one text contained in the other, keep the longer page. The window grows to ±5 pages, and the comparison only runs when the two lengths are within 10% of each other.

14. Recording

The database row is unchanged: `blank_junk_flag`, `junk_subtype`, `confidence` (≤ 0.60 when uncertain) and `reason` (the deciding rule, e.g. `R9 vote JKK row 4 -> Main (veto: clinical)`). No schema change.

Every ballot is staged on disk beside the key/value extraction's `extraction.json`, at `<chart>/staging/blank_junk.json`. Each pass replaces its own entries and keeps the other pass's:


```
        {"version": 1, "chart_name": "53688890", "updated_at": "2026-10-15T10:12:03+00:00",
         "pages": {"24.jpg": {"pass_2": {
           "pass": 2, "rule": "R9",
           "bert":  {"vote": "JUNK", "strength": "strong", "p_junk": 0.98, "subtype": "Record
Request/Transmittal",
                     "label": "JUNK_RECORD_REQUEST"},
           "tfidf": {"vote": "KEEP", "strength": "strong", "p_junk": 0.0},
           "kw":    {"vote": "KEEP", "strength": "weak", "p_junk": 0.0, "clinical": 5, "junk": 0,
                     "demographic": 1, "junk_group": null, "hits": {"clinical.section_headings":
["PLAN"]}},
           "ballots": "JKK", "row": 4, "r4_skipped": "clinical", "uncertain": true,
           "visual_source": "none", "signature_source": "kv_extract", "hw_area": "none",
           "short_page": {"content_bottom": 0.415},
           "verdict": "Main", "confidence": 0.6, "reason": "R9 vote JKK row 4 -> Main"}}}}
```


`visual_source` is `none` when the page-tag model did not tag the page, `signature_source` is `canon_only` when the KV signature output was missing, `short_page` is `no_geometry` when the page had no word boxes, and a model that was not loaded shows `"note": "unavailable"` on its ballot. A duplicate keeps the votes of its verdict, plus a `duplicate` entry.

15. Checked on the local charts (2026-10-15)

Implemented directly, without the shadow-mode step. Run on the three local charts that have Final OCR (84 pages), against the old verdicts in their `junk.csv`:

Result	Pages
Same verdict as before	80
R4 junk, old Main — header / footer only (`Printed … Pg 8 of 10`)	1 (53688890 p28)
R4 junk, old Main — tail of a note (care team, confidentiality footer)	1 (53688890 p16)
Patient-education handouts (fall prevention, first-aid kit): R4 junk until the `patient_education` canon group was added; now Main (clinical 6, confidence 0.60)	2 (60306165 p27, p29)
Face sheet and BCG administration page	kept after R4 was narrowed (vote, confidence 0.60)

Still to do: compare against `page_ground_truth` (`blank_page`, `junk_page`) per voter and per rule from `staging/blank_junk.json`, with the number that matters being clinical pages marked junk; and check R5 by hand on charts run with the page-tag model.

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16. Open points

1. `advantmed` is listed as an ROI vendor (its own retrieval cover sheets). Remove it if Advantmed letterhead appears on clinical pages.
2. Should R5 also run in pass 1 using ink alone? It would save stage-5 cost on short pages, but it would lose the signature exemption.
3. DistilBERT force (R4) keeps signed pages. Should a signed page above 85% be junked too? Should R4 also run in pass 1, saving the stage-5 bill but before Provider Signature is known?